

FACS Analysis of Mitochondrial Membrane Potential and CRISPR Screen Read-out via Amplicon Sequencing

For our mitochondrial assays we used a nearly identical library (same backbone, guide density, and control set) screened exclusively in the H1 inducible Cas9 (H1-iCas9) hPSC line. Mitochondrial inner membrane potential ($\Delta\psi_m$) was measured in H1-iCas9, following differentiation to NPCs or iGut on day 21. Cells were harvested, counted, and aliquoted at 1×10^6 cells per sample. JC-1 dye (MitoProbe™ JC-1 Assay Kit; Invitrogen #M34152) was dissolved in DMSO at a stock concentration of 200 μ M and added to each sample to achieve a final concentration of 2 μ M, then incubated for 30 min at 37 °C in 5% CO₂. A 50 μ M CCCP control was included to induce complete mitochondrial depolarization. After staining, cells were washed once in their respective culture medium, resuspended in FACS buffer (Invitrogen eBioscience Staining Buffer #00422226), and analyzed immediately on a Thermo Fisher “Bigfoot” spectral cell sorter using 488 nm excitation with 525/50 nm (FITC) and 585/40 nm (PE) emission filters. Debris and doublets were excluded by forward/side scatter gating, and CCCP-treated samples were used to define FITC and PE gates. Approximately 1×10^6 events per sample were recorded. Cells were then pelleted ($300 \times g$, 5 min) and genomic DNA extracted using the Qiagen DNeasy Blood & Tissue Kit (#69504).

UMI-tagged amplicon libraries were generated in three PCR steps. In PCR-1, genomic DNA was amplified with Platinum™ II Hot-Start PCR Master Mix (Invitrogen, #14000012) and UMI-containing primers (Forward: 5'-ACACTCTTTCCCTACACGACGCTCTTCCGATCTACGTGACGTAGAAAGTAATAATTTCTTGGGT-3'; Reverse: 5'-GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCTN(25)NNNNNNNNNACTCGGTGCCACTTTTCAA-3') under the following conditions: 94 °C for 2 min; 4–6 cycles of 98 °C for 5 s, 60 °C for 15 s, 60 °C for 30 s. The resulting ~180 bp products were purified and concentrated using the Zymo DNA Clean & Concentrator-5 kit (#D4013) and eluted in 10 μ L nuclease-free water (Thermo Fisher #AM9938). In PCR-2, purified product was amplified with adaptor primers (Forward: 5'-ACACTCTTTCCCTACACGACGCTCTTCCGATCT-3'; Reverse: 5'-GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT-3') for 22 cycles under identical cycling conditions in a ~20 μ L reaction. A seven-cycle indexing PCR (PCR-3) was performed by the sequencing facility Yale Center for Genome Analysis (YCGA) prior to sequencing. Final libraries were sequenced on an Illumina NovaSeq platform (paired-end 150 bp, 5 million reads per sample).

Flanking sequences on both sides of each gRNA were trimmed using BBDuk, and reads were then mapped to gRNA reference sequences and counted using MAGeCK. Raw counts for each gRNA were normalized to counts of scrambled gRNA. Abundance of each target gene was then calculated by summing of all gRNAs targeting that gene. Log₂-transformed fold changes of gRNA-targets abundance were compared between PE-high samples and FITC-high samples.